

Supplemental Data

Molecules are indexed by a unique identifier (RedJade Code) allowing for reproduction of most results shown here. Information required to identify chemical structures will be provided at <https://github.com/osmoai/publications>.

Data S1. Metadata for 400 molecules in the validation dataset.
Columns in the dataset are defined as follows:

RedJade Code	Internal anonymizing tracking number for panelist responses [PRIMARY KEY]
Solvent	Diluting solvent, if needed for safety or for intensity balancing
Final []	Concentration of molecule (w/w) in final sample
GC-O no of beads used	Number of viscopearls with absorbed odorant used for headspace extraction prior to GC-O analysis
GC-O incubation and extraction temperature	GC-O methodological details
GC-O incubation time	
GC-O extraction time	
GC-O result	Verdict from GC-O analysis
GC-O contaminant, if identified	Canonical SMILES of the causal contaminant, if one was successfully identified
Impact on GNN performance	Whether the GC-O result had a good, bad, neutral, or unknown effect on GNN's prediction performance.
Disqualification reason	Reason for disqualification. If blank, molecule was retained for analysis
Selection reason	Original selection criteria. Molecules were predicted by the GNN or Random Forest model to have an odor prediction above some threshold despite structural dissimilarity to known instances of that odor class, or to have an odor prediction below some threshold, despite structural similarity to known instances of that odor class.

Data S2. Panelist evaluations of 20 common odorants. Prospective panelists gave RATA ratings using the 55-word lexicon for 20 common odorants; panelists with a raw test-retest correlation greater than 0.35 were invited to join the panel.

Data S3. Panelist evaluations of 397 novel odorants. Between 15 and 18 panelists rated intensity and pleasantness and gave RATA ratings using the 55-word lexicon for each molecule.

Data S4. Odor attribute predictions on 397 molecules by a random forest model trained on GS-LF datasets.

Data S5. Odor attribute predictions on 397 molecules by a graph neural network model trained on GS-LF datasets. Final layer.

Data S6. Two dimensional coordinates (for visualization) of the high-dimensional graph neural network embeddings on 397 molecules. Penultimate layer.

Data S7. Correspondence table between internal odorant identifiers and chemical structures.